

Qualification Pack

Data Analysis Associate

QP Code: NIE/ITS/Q1101

Version: 1.0

NSQF Level: 4

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NIE/ITS/Q1101: Data Analysis Associate

Brief Job Description

It is meant to inculcate the basic skills of data handling, processing and visualization. The trained learner will be able to view data in summary and aggregated form, he/she would be able to formulate equation for prediction. inculcate the basic skills of Hadoop framework and programming experience of core Java.

Personal Attributes

should have strong analytical and problem-solving skills with attention to detail. They must be proficient in data handling, interpretation, and visualization using relevant tools. Good communication skills are essential for presenting insights effectively. Additionally, they should be adaptable, ethical, and committed to data accuracy and confidentiality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
2. [NIE/SSC/N1114: Implementation of Data Analysis Project--](#)
3. [NIE/ITS/N1101: Perform basic calculation using spreadsheet--](#)
4. [NIE/ITS/N1102: Manage structured data--](#)
5. [NIE/ITS/N1103: Analyse data using spreadsheet tool--](#)
6. [NIE/ITS/N1104: Manage data in Open source tool--](#)
7. [NIE/ITS/N1106: Visualize data graphically--](#)
8. [NIE/ITS/N1117: Installation of Hadoop Framework and Java Programming--](#)
9. [NIE/ITS/N1118: Manage big data using hadoop--](#)
10. [NIE/ITS/N1119: Implementation of use cases of Data Analytics--](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	DATA SCIENCE & ENGINEERING

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Occupation	DATA ANALYTICS
Country	India
NSQF Level	4
Credits	15
Aligned to NCO/ISCO/ISIC Code	2120.0501
Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of 3-year diploma (after 10th) and pursuing regular diploma with NA of experience OR 10th grade pass plus 2-year NTC with NA of experience OR 10th grade pass plus 1-year NTC/ NAC with NA of experience OR 10th grade pass and pursuing continuous schooling with NA of experience OR Not Applicable (Data nalysis Assistant NOS from 1-6 will be Exempted Previous NSQF Qualification of Level 3.) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	30/06/2026
NSQC Approval Date	29/03/2023
Version	1.0
Reference code on NQR	QG-04-IT-00348-2023-V1-NIELIT
NQR Version	1

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

NIE/SSC/N1114: Implementation of Data Analysis Project--

Description

This National Occupational Standard (NOS) focuses on applying data analysis skills through a structured project. It emphasizes practical experience in handling real-world data, utilizing various analytical tools, and implementing data-driven solutions. Key competencies include the ability to clean, process, and visualize data effectively, ensuring data accuracy and integrity. The project encourages critical thinking, problem-solving, and collaboration to derive meaningful insights. Additionally, it highlights the importance of working in a professional environment, adhering to industry best practices, and staying updated with emerging trends in data analytics.

Scope

The scope covers the following :

- The National Occupational Standard (NOS) for Data Analysis defines key competencies in data handling, processing, and visualization. It covers database management, statistical analysis, big data tools, and real-world project implementation. The NOS ensures industry relevance by integrating employability skills and aligning with emerging trends in data analytics, preparing professionals for data-driven decision-making roles.

Elements and Performance Criteria

Design and Implementation of Data Analysis Associate

To be competent, the user/individual on the job must be able to:

- PC1.** Apply Pivot Tables, statistical analysis, and advanced Excel functions for insights.
- PC2.** Utilize MapReduce and Hive for efficient data storage, processing, and analysis.
- PC3.** Manage structured data using SQL, database keys, and integrity constraints for accurate data handling.
- PC4.** Create data visualizations using Tableau, connecting to multiple data sources for insightful analysis.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Acquire skills in SQL, database keys, integrity rules, data migration, and cleaning.
- KU2.** Develop proficiency in Excel functions, Pivot Tables, statistical techniques, regression analysis, and data validation.
- KU3.** Gain knowledge of Hadoop, HDFS, MapReduce, Hive, and their role in big data analytics.
- KU4.** Acquire skills in Tableau for data connection, formatting, sorting, filtering, and interactive dashboard creation.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** Analyze and interpret data using statistical techniques and visualization tools for decision-making.
- GS2.** Effectively communicate data insights and findings to stakeholders using reports and presentations.
- GS3.** Collaborate with cross-functional teams to solve data-related problems and implement analytical solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design and Implementation of Data Analysis Associate</i>	-	-	-	30
PC1. Apply Pivot Tables, statistical analysis, and advanced Excel functions for insights.	-	-	-	-
PC2. Utilize MapReduce and Hive for efficient data storage, processing, and analysis.	-	-	-	-
PC3. Manage structured data using SQL, database keys, and integrity constraints for accurate data handling.	-	-	-	-
PC4. Create data visualizations using Tableau, connecting to multiple data sources for insightful analysis.	-	-	-	-
NOS Total	-	-	-	30

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1114
NOS Name	Implementation of Data Analysis Project--
Sector	IT-ITeS
Sub-Sector	
Occupation	Data Analytics
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1101: Perform basic calculation using spreadsheet--

Description

Spreadsheet software facilitates the creation, organization, and storage of multiple worksheets within workbooks, enabling users to manage data efficiently. It also allows for the customization of text formats and alignment, supports basic functions and formulas for data manipulation, enables sorting for data organization, and facilitates the creation of graphs for visual data representation and analysis.

Scope

The scope covers the following :

- Spreadsheet features, Creating and saving worksheet and workbook layouts, learning basic functions and formulas, sorting and Graphs

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.**
- Creating and saving worksheet and workbook:
 - Understanding of file management and organization in a spreadsheet environment.
 - Proficiency in creating, saving, and managing multiple worksheets within a workbook.
- PC2.**
- Knowledge of formatting tools to enhance data presentation and readability.
 - Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.
- PC3.**
- Proficiency in using basic mathematical and logical functions for data manipulation.
 - Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.
 - Understanding the importance of data organization and its impact on data analysis.
 - Proficiency in sorting data based on specific criteria for effective data management and interpretation.
- PC4.**
- Ability to interpret and present data visually through graphs and charts.
 - Understanding the use of graphical representations for effective communication and data analysis.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the importance of data organization and its impact on data analysis. Proficiency in sorting data based on specific criteria for effective data management and interpretation.
- KU2.** Proficiency in using basic mathematical and logical functions for data manipulation. Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.
- KU3.** Knowledge of formatting tools to enhance data presentation and readability. Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.

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- KU4.** Understanding of file management and organization in a spreadsheet environment.
Proficiency in creating, saving, and managing multiple worksheets within a workbook.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
• Effective management of data, files, and workbooks. Analytical Skills: Ability to analyze and interpret data using functions and formulas. Attention to Detail: Ensuring accurate data entry, formatting, and alignment. Problem-Solving: Identifying and resolving issues related to data organization and presentation. Communication Skills: Effectively conveying insights through well-designed graphs and charts.
- GS2.** • Professional Skills:
• Data Management: Efficiently handling large datasets and maintaining data integrity. Technical Proficiency: Mastery of spreadsheet software tools and functions for optimized usage. Presentation Skills: Creating visually appealing and informative data presentations for professional settings. Critical Thinking: Evaluating data trends and patterns to make informed business decisions. Collaboration: Working with teams to present and analyze data effectively for reports.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	23	10	-	-
PC1. - Creating and saving worksheet and workbook: - Understanding of file management and organization in a spreadsheet environment. - Proficiency in creating, saving, and managing multiple worksheets within a workbook.	-	-	-	-
PC2. - Knowledge of formatting tools to enhance data presentation and readability. - Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.	-	-	-	-
PC3. - Proficiency in using basic mathematical and logical functions for data manipulation. - Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet. - Understanding the importance of data organization and its impact on data analysis. - Proficiency in sorting data based on specific criteria for effective data management and interpretation.	-	-	-	-
PC4. - Ability to interpret and present data visually through graphs and charts. - Understanding the use of graphical representations for effective communication and data analysis.	-	-	-	-
NOS Total	23	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1101
NOS Name	Perform basic calculation using spreadsheet--
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1101: Perform basic calculation using spreadsheet.

Description

Spreadsheet software facilitates the creation, organization, and storage of multiple worksheets within workbooks, enabling users to manage data efficiently. It also allows for the customization of text formats and alignment, supports basic functions and formulas for data manipulation, enables sorting for data organization, and facilitates the creation of graphs for visual data representation and analysis.

Scope

The scope covers the following :

- Spreadsheet features, Creating and saving worksheet and workbook layouts, learning basic functions and formulas, sorting and Graphs

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.**
- Creating and saving worksheet and workbook:
 - Understanding of file management and organization in a spreadsheet environment.
 - Proficiency in creating, saving, and managing multiple worksheets within a workbook.
- PC2.**
- Knowledge of formatting tools to enhance data presentation and readability.
 - Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.
- PC3.**
- Proficiency in using basic mathematical and logical functions for data manipulation.
 - Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.
 - Understanding the importance of data organization and its impact on data analysis.
 - Proficiency in sorting data based on specific criteria for effective data management and interpretation.
- PC4.**
- Ability to interpret and present data visually through graphs and charts.
 - Understanding the use of graphical representations for effective communication and data analysis.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the importance of data organization and its impact on data analysis. Proficiency in sorting data based on specific criteria for effective data management and interpretation.
- KU2.** Proficiency in using basic mathematical and logical functions for data manipulation. Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.
- KU3.** Knowledge of formatting tools to enhance data presentation and readability. Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.

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- KU4.** Understanding of file management and organization in a spreadsheet environment.
Proficiency in creating, saving, and managing multiple worksheets within a workbook.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
• Effective management of data, files, and workbooks. Analytical Skills: Ability to analyze and interpret data using functions and formulas. Attention to Detail: Ensuring accurate data entry, formatting, and alignment. Problem-Solving: Identifying and resolving issues related to data organization and presentation. Communication Skills: Effectively conveying insights through well-designed graphs and charts.
- GS2.** • Professional Skills:
• Data Management: Efficiently handling large datasets and maintaining data integrity. Technical Proficiency: Mastery of spreadsheet software tools and functions for optimized usage. Presentation Skills: Creating visually appealing and informative data presentations for professional settings. Critical Thinking: Evaluating data trends and patterns to make informed business decisions. Collaboration: Working with teams to present and analyze data effectively for reports.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	13	6	-	-
PC1. - Creating and saving worksheet and workbook: - Understanding of file management and organization in a spreadsheet environment. - Proficiency in creating, saving, and managing multiple worksheets within a workbook.	13	6	-	-
PC2. - Knowledge of formatting tools to enhance data presentation and readability. - Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.	-	-	-	-
PC3. - Proficiency in using basic mathematical and logical functions for data manipulation. - Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet. - Understanding the importance of data organization and its impact on data analysis. - Proficiency in sorting data based on specific criteria for effective data management and interpretation.	-	-	-	-
PC4. - Ability to interpret and present data visually through graphs and charts. - Understanding the use of graphical representations for effective communication and data analysis.	-	-	-	-
NOS Total	13	6	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1101
NOS Name	Perform basic calculation using spreadsheet.
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4.5
Credits	1
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1102: Manage structured data--

Description

Introduction to database , advantages of DBMS, Concept of keys: candidate key, primary key, alternate key, Foreign key, Fundamental integrity rules: entity integrity, referential integrity. Entity-relationship model .ER-diagram, SQL, Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations. Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations.

Scope

The scope covers the following :

- It encompasses the foundational understanding of databases, the benefits of Database Management Systems (DBMS), and an in-depth exploration of key concepts including candidate keys, primary keys, alternate keys, foreign keys, entity integrity, and referential integrity. Additionally, it involves the application of the Entity-Relationship model for database design, the mastery of SQL for efficient data management, and a holistic view of Knowledge Discovery in Databases (KDD), data mining techniques, and the architecture of data warehouses. The scope further extends to encompass the methodologies of migrating data to data warehouses, the pivotal process of data cleaning, and the application of aggregation operations to ensure data accuracy and integrity for effective decision-making and analysis.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding the fundamentals of databases, their types, and their importance in information management systems.
- PC2.** Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.
- PC3.** Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.
- PC4.**
 - Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets. Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis.
 - Migrating Data to Data

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** Individual learn about Concepts of Database and Its Advantages
- KU2.** Concept of Keys, Fundamental Integrity Rules, Entity-Relationship Model (ER Diagram):
- KU3.** Mastering the Structured Query Language (SQL), Knowledge Discovery in Databases (KDD),
- KU4.** Learning concepts of Data Mining and Data Warehouse

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
 - Time Management: Efficiently managing tasks and adhering to project timelines, especially during data migration and cleaning operations. Resource Management: Effectively utilizing available tools and resources for database management and data mining processes. Data Organization: Structuring data in a systematic manner to ensure efficient access and retrieval during analysis and reporting.
 - Project Planning: Strategically planning and implementing data warehouse initiatives, taking into account data migration, cleaning, and aggregation operations. Detail Orientation: Paying close attention to data details to maintain accuracy and integrity throughout the data management and analysis processes.
- GS2.** • Professional Skills:
 - Technical Proficiency: Demonstrating an advanced understanding of database management systems, SQL querying, and data mining techniques for proficient handling and analysis of large datasets.
 - Problem-Solving: Resolving complex database-related issues and challenges during data migration, cleaning, and aggregation, ensuring optimal data quality and integrity.
 - Communication Skills: Articulating complex technical concepts and insights to both technical and non-technical stakeholders in a clear and concise manner. Collaboration: Working effectively with multidisciplinary teams, including data analysts, IT professionals, and stakeholders, to ensure seamless data management and analysis processes. Critical Thinking: Evaluating data trends and patterns critically to make informed decisions during data mining and knowledge discovery processes.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	24	10	-	-
PC1. Understanding the fundamentals of databases, their types, and their importance in information management systems.	-	-	-	-
PC2. Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.	-	-	-	-
PC3. Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.	-	-	-	-
PC4. - Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets. - Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis. - Migrating Data to Data	-	-	-	-
NOS Total	24	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1102
NOS Name	Manage structured data--
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1102: Manage structured data.

Description

Introduction to database , advantages of DBMS, Concept of keys: candidate key, primary key, alternate key, Foreign key, Fundamental integrity rules: entity integrity, referential integrity. Entity-relationship model .ER-diagram, SQL, Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations. Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations.

Scope

The scope covers the following :

- It encompasses the foundational understanding of databases, the benefits of Database Management Systems (DBMS), and an in-depth exploration of key concepts including candidate keys, primary keys, alternate keys, foreign keys, entity integrity, and referential integrity. Additionally, it involves the application of the Entity-Relationship model for database design, the mastery of SQL for efficient data management, and a holistic view of Knowledge Discovery in Databases (KDD), data mining techniques, and the architecture of data warehouses. The scope further extends to encompass the methodologies of migrating data to data warehouses, the pivotal process of data cleaning, and the application of aggregation operations to ensure data accuracy and integrity for effective decision-making and analysis.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding the fundamentals of databases, their types, and their importance in information management systems.
- PC2.** Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.
- PC3.** Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.
- PC4.**
 - Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets. Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis.
 - Migrating Data to Data

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Individual learn about Concepts of Database and Its Advantages
- KU2.** Concept of Keys, Fundamental Integrity Rules, Entity-Relationship Model (ER Diagram):
- KU3.** Mastering the Structured Query Language (SQL), Knowledge Discovery in Databases (KDD),
- KU4.** Learning concepts of Data Mining and Data Warehouse

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
 - Time Management: Efficiently managing tasks and adhering to project timelines, especially during data migration and cleaning operations. Resource Management: Effectively utilizing available tools and resources for database management and data mining processes. Data Organization: Structuring data in a systematic manner to ensure efficient access and retrieval during analysis and reporting.
 - Project Planning: Strategically planning and implementing data warehouse initiatives, taking into account data migration, cleaning, and aggregation operations. Detail Orientation: Paying close attention to data details to maintain accuracy and integrity throughout the data management and analysis processes.
- GS2.** • Professional Skills:
 - Technical Proficiency: Demonstrating an advanced understanding of database management systems, SQL querying, and data mining techniques for proficient handling and analysis of large datasets.
 - Problem-Solving: Resolving complex database-related issues and challenges during data migration, cleaning, and aggregation, ensuring optimal data quality and integrity.
 - Communication Skills: Articulating complex technical concepts and insights to both technical and non-technical stakeholders in a clear and concise manner. Collaboration: Working effectively with multidisciplinary teams, including data analysts, IT professionals, and stakeholders, to ensure seamless data management and analysis processes. Critical Thinking: Evaluating data trends and patterns critically to make informed decisions during data mining and knowledge discovery processes.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	12	6	-	-
PC1. Understanding the fundamentals of databases, their types, and their importance in information management systems.	12	6	-	-
PC2. Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.	-	-	-	-
PC3. Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.	-	-	-	-
PC4. - Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets. - Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis. - Migrating Data to Data	-	-	-	-
NOS Total	12	6	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1102
NOS Name	Manage structured data.
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4.5
Credits	1
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1103: Analyse data using spreadsheet tool--

Description

Student will learn the key elements of data manipulation and analysis within spreadsheet software. Data entails the information input, while organizing it into tables and formatting involves structuring and visually enhancing the presentation. Conditional formatting enables users to dynamically format cells based on specific conditions, advanced charting aids in creating comprehensive visual representations, and pivot tables provide an effective means of data summarization. Data validation ensures the accuracy and reliability of data input, while filtering allows for focused data analysis by selectively displaying subsets of information that meet specified criteria.

Scope

The scope covers the following :

- The scope involves comprehensive training in data handling within spreadsheets, focusing on data input, organization, and presentation through tables and formatting. It also includes an in-depth understanding of conditional formatting techniques, advanced charting methodologies for sophisticated data visualization, and the proficient use of pivot tables for efficient data summarization and analysis. Additionally, the scope entails expertise in implementing data validation measures to ensure data accuracy, as well as utilizing filtering techniques to extract specific data subsets, thereby facilitating effective data analysis and decision-making.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding data entry and storage within spreadsheets. Learning to create and manage tables for organized data representation. Mastering formatting tools to enhance the visual presentation of data.
- PC2.** Exploring the application of formatting rules based on specific conditions. Understanding how to dynamically highlight and format data for easy analysis. Learning to use color scales, data bars, and icon sets to represent data variations effectively.
- PC3.** Gaining proficiency in creating a variety of complex charts to visualize data trends and patterns. Understanding the use of trendlines, dual axes, and custom data labels for comprehensive data representation. Learning to create combination charts to display multiple data sets in a single graph. Learning how to summarize and analyze large datasets efficiently. Understanding the process of data aggregation and dynamic restructuring for insightful analysis.
- PC4.** Implementing data validation rules to ensure accurate and consistent data entry. Understanding the use of input messages and error alerts for guiding data input. Learning to restrict data input based on predefined criteria for data integrity. Understanding how to filter data based on specific criteria to focus on relevant information. Exploring the use of filter views to create customized views of data subsets. Learning to apply multiple filters for complex data analysis and visualization.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Candidate will gain knowledge of Data manipulation inside ant Table, Formatting and conditional formatting
- KU2.** Exploring the application of formatting rules based on specific conditions. Understanding how to dynamically highlight and format data for easy analysis.
- KU3.** Candidate will gain knowledge of Advance Charting and use of Pivot Table
- KU4.** Implementing data validation rules to ensure accurate and consistent data entry. Understanding the use of input messages and error alerts for guiding data input. Exploring the use of filter views to create customized views of data subsets.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
 - he proficient application of the aforementioned spreadsheet functionalities necessitates strong organizational skills, including adept time management for efficient handling of data tasks, meticulous attention to detail to ensure accurate data organization and analysis, and effective task prioritization to manage various aspects of data manipulation. Additionally, the ability to strategically plan and execute data management projects, coupled with proficient problem-solving capabilities and clear communication skills for conveying data insights, are crucial for successful implementation and interpretation of spreadsheet data
- GS2.** • Professional Skills:
 - Proficient application of these features in a professional context necessitates a strong data analysis proficiency, along with technical expertise in spreadsheet software and related tools for efficient data manipulation. Critical thinking skills are essential for insightful data interpretation, while maintaining data integrity through meticulous validation and filtering procedures is crucial. Effective problem-solving abilities enable the resolution of complex data-related challenges, and adaptability to new techniques ensures the incorporation of the latest advancements in spreadsheet technology. Additionally, collaborative aptitude facilitates effective teamwork in analyzing and interpreting data for informed decision-making and problem-solving in a professional environment.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	35	10	-	-
PC1. Understanding data entry and storage within spreadsheets. Learning to create and manage tables for organized data representation. Mastering formatting tools to enhance the visual presentation of data.	-	-	-	-
PC2. Exploring the application of formatting rules based on specific conditions. Understanding how to dynamically highlight and format data for easy analysis. Learning to use color scales, data bars, and icon sets to represent data variations effectively.	-	-	-	-
PC3. Gaining proficiency in creating a variety of complex charts to visualize data trends and patterns. Understanding the use of trendlines, dual axes, and custom data labels for comprehensive data representation. Learning to create combination charts to display multiple data sets in a single graph. Learning how to summarize and analyze large datasets efficiently. Understanding the process of data aggregation and dynamic restructuring for insightful analysis.	-	-	-	-
PC4. Implementing data validation rules to ensure accurate and consistent data entry. Understanding the use of input messages and error alerts for guiding data input. Learning to restrict data input based on predefined criteria for data integrity. Understanding how to filter data based on specific criteria to focus on relevant information. Exploring the use of filter views to create customized views of data subsets. Learning to apply multiple filters for complex data analysis and visualization.	-	-	-	-
NOS Total	35	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1103
NOS Name	Analyse data using spreadsheet tool--
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1103: Analyse data using spreadsheet tool.

Description

Student will learn the key elements of data manipulation and analysis within spreadsheet software. Data entails the information input, while organizing it into tables and formatting involves structuring and visually enhancing the presentation. Conditional formatting enables users to dynamically format cells based on specific conditions, advanced charting aids in creating comprehensive visual representations, and pivot tables provide an effective means of data summarization. Data validation ensures the accuracy and reliability of data input, while filtering allows for focused data analysis by selectively displaying subsets of information that meet specified criteria.

Scope

The scope covers the following :

- The scope involves comprehensive training in data handling within spreadsheets, focusing on data input, organization, and presentation through tables and formatting. It also includes an in-depth understanding of conditional formatting techniques, advanced charting methodologies for sophisticated data visualization, and the proficient use of pivot tables for efficient data summarization and analysis. Additionally, the scope entails expertise in implementing data validation measures to ensure data accuracy, as well as utilizing filtering techniques to extract specific data subsets, thereby facilitating effective data analysis and decision-making.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding data entry and storage within spreadsheets. Learning to create and manage tables for organized data representation. Mastering formatting tools to enhance the visual presentation of data.
- PC2.** Exploring the application of formatting rules based on specific conditions. Understanding how to dynamically highlight and format data for easy analysis. Learning to use color scales, data bars, and icon sets to represent data variations effectively.
- PC3.** Gaining proficiency in creating a variety of complex charts to visualize data trends and patterns. Understanding the use of trendlines, dual axes, and custom data labels for comprehensive data representation. Learning to create combination charts to display multiple data sets in a single graph. Learning how to summarize and analyze large datasets efficiently. Understanding the process of data aggregation and dynamic restructuring for insightful analysis.
- PC4.** Implementing data validation rules to ensure accurate and consistent data entry. Understanding the use of input messages and error alerts for guiding data input. Learning to restrict data input based on predefined criteria for data integrity. Understanding how to filter data based on specific criteria to focus on relevant information. Exploring the use of filter views to create customized views of data subsets. Learning to apply multiple filters for complex data analysis and visualization.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Candidate will gain knowledge of Data manipulation inside ant Table, Formatting and conditional formatting
- KU2.** Exploring the application of formatting rules based on specific conditions. Understanding how to dynamically highlight and format data for easy analysis.
- KU3.** Candidate will gain knowledge of Advance Charting and use of Pivot Table
- KU4.** Implementing data validation rules to ensure accurate and consistent data entry. Understanding the use of input messages and error alerts for guiding data input. Exploring the use of filter views to create customized views of data subsets.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
 - he proficient application of the aforementioned spreadsheet functionalities necessitates strong organizational skills, including adept time management for efficient handling of data tasks, meticulous attention to detail to ensure accurate data organization and analysis, and effective task prioritization to manage various aspects of data manipulation. Additionally, the ability to strategically plan and execute data management projects, coupled with proficient problem-solving capabilities and clear communication skills for conveying data insights, are crucial for successful implementation and interpretation of spreadsheet data
- GS2.** • Professional Skills:
 - Proficient application of these features in a professional context necessitates a strong data analysis proficiency, along with technical expertise in spreadsheet software and related tools for efficient data manipulation. Critical thinking skills are essential for insightful data interpretation, while maintaining data integrity through meticulous validation and filtering procedures is crucial. Effective problem-solving abilities enable the resolution of complex data-related challenges, and adaptability to new techniques ensures the incorporation of the latest advancements in spreadsheet technology. Additionally, collaborative aptitude facilitates effective teamwork in analyzing and interpreting data for informed decision-making and problem-solving in a professional environment.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	37	18	-	-
PC1. Understanding data entry and storage within spreadsheets. Learning to create and manage tables for organized data representation. Mastering formatting tools to enhance the visual presentation of data.	37	18	-	-
PC2. Exploring the application of formatting rules based on specific conditions. Understanding how to dynamically highlight and format data for easy analysis. Learning to use color scales, data bars, and icon sets to represent data variations effectively.	-	-	-	-
PC3. Gaining proficiency in creating a variety of complex charts to visualize data trends and patterns. Understanding the use of trendlines, dual axes, and custom data labels for comprehensive data representation. Learning to create combination charts to display multiple data sets in a single graph. Learning how to summarize and analyze large datasets efficiently. Understanding the process of data aggregation and dynamic restructuring for insightful analysis.	-	-	-	-
PC4. Implementing data validation rules to ensure accurate and consistent data entry. Understanding the use of input messages and error alerts for guiding data input. Learning to restrict data input based on predefined criteria for data integrity. Understanding how to filter data based on specific criteria to focus on relevant information. Exploring the use of filter views to create customized views of data subsets. Learning to apply multiple filters for complex data analysis and visualization.	-	-	-	-
NOS Total	37	18	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1103
NOS Name	Analyse data using spreadsheet tool.
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4.5
Credits	3
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1104: Manage data in Open source tool--

Description

It is an introduction to Virtual Machines, covering the creation and configuration processes. It further delves into the history and benefits of Linux, exploring the various Linux distributions and focusing on Ubuntu. The scope includes a step-by-step guide to installing Ubuntu, familiarizing users with startup procedures, logging in, and navigating the desktop environment. Additionally, it involves an in-depth understanding of the Linux file system, covering concepts such as file management, real and virtual files, mounting, file searches, file size, and viewing text files. Moreover, the scope encompasses practical guidance on using command-line text editors, creating, searching, and comparing text files, as well as managing files through copying, moving, and other relevant operations within the Linux environment.

Scope

The scope covers the following :

- Comprehensive understanding of virtualization concepts with a focus on Virtual Machines, covering their introduction, creation, and configuration. Additionally, it entails an exploration of the history of Linux, its benefits, and the various Linux distributions, with a specific emphasis on Ubuntu. The scope involves a detailed walkthrough of the installation process for Ubuntu, guiding users through startup procedures, logging in, and desktop navigation. It also encompasses an in-depth exploration of the Linux file system, covering fundamental concepts such as file management, file system structure, and various file operations. Furthermore, the scope includes practical training in using command-line text editors, file searching, comparison, and manipulation, enabling users to effectively manage files and directories within the Linux environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Introduction to Virtual Machine, Creating and Configuring Virtual Machine, Linux History

PC2. Benefits of Linux, Different Flavors of Linux, Introducing Ubuntu

PC3. Installing Ubuntu: Starting Up, Logging in, Exploring the Desktop, Ubuntu Basics

PC4. Browsing the File System, Understanding File System Concept, Managing Files: Real and Virtual Files, Mounting, File Searches, File Size, File Space Viewing, Text Files Viewing, Using a Command, Line Text Editor, Creating Files, Searching through Files, Comparing Text Files, Copying, Moving, Managing Files

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Virtual Machines: Knowledge of virtualization technology and software tools for creating and configuring virtual machines. Linux History and Benefits: Understanding the origins, development, and advantages of the Linux operating system over other operating systems.
- KU2.** Linux Distributions: Awareness of various Linux distributions, their features, and use cases, with a specific focus on Ubuntu.

Qualification Pack

- KU3.** Ubuntu Installation and Basics: Familiarity with the installation process and basic operations within the Ubuntu Linux environment, including desktop navigation and file system management.
- KU4.** File System Operations: Understanding the structure and management of file systems in Linux, including file browsing, searching, size estimation, and text file manipulation using command-line text editors. File Management: Knowledge of file management operations such as copying, moving, and comparing files, as well as handling real and virtual files and mounting external file systems.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
- The proficient execution of the outlined tasks necessitates adept organizational skills, including effective time management for efficient handling of tasks such as virtual machine configuration, Linux installation, and file management within the Ubuntu environment. Furthermore, meticulous attention to detail is crucial during file system navigation, file searches, and text file manipulation, ensuring accurate and precise data handling. Additionally, proficient task prioritization and problem-solving abilities are essential for addressing any challenges that may arise during the installation process and for maintaining an organized approach to data and file management operations.
- GS2.** • Professional Skills:
- Proficiency in the aforementioned tasks requires a combination of professional skills, including a strong grasp of virtualization technologies and Linux operating systems, effective problem-solving abilities to address technical challenges during virtual machine configuration and Linux installation, and meticulous attention to detail for accurate file system management and file manipulation. Additionally, effective communication skills are essential for documenting procedures and troubleshooting steps, while adaptability and continuous learning ensure the ability to stay updated with the latest developments in virtualization and Linux environments.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	18	15	-	-
PC1. Introduction to Virtual Machine, Creating and Configuring Virtual Machine, Linux History	-	-	-	-
PC2. Benefits of Linux, Different Flavors of Linux, Introducing Ubuntu	-	-	-	-
PC3. Installing Ubuntu: Starting Up, Logging in, Exploring the Desktop, Ubuntu Basics	-	-	-	-
PC4. Browsing the File System, Understanding File System Concept, Managing Files: Real and Virtual Files, Mounting, File Searches, File Size, File Space Viewing, Text Files Viewing, Using a Command , Line Text Editor, Creating Files, Searching through Files, Comparing Text Files, Copying, Moving, Managing Files	-	-	-	-
NOS Total	18	15	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1104
NOS Name	Manage data in Open source tool--
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1104: Manage data in Open source tool.

Description

It is an introduction to Virtual Machines, covering the creation and configuration processes. It further delves into the history and benefits of Linux, exploring the various Linux distributions and focusing on Ubuntu. The scope includes a step-by-step guide to installing Ubuntu, familiarizing users with startup procedures, logging in, and navigating the desktop environment. Additionally, it involves an in-depth understanding of the Linux file system, covering concepts such as file management, real and virtual files, mounting, file searches, file size, and viewing text files. Moreover, the scope encompasses practical guidance on using command-line text editors, creating, searching, and comparing text files, as well as managing files through copying, moving, and other relevant operations within the Linux environment.

Scope

The scope covers the following :

- Comprehensive understanding of virtualization concepts with a focus on Virtual Machines, covering their introduction, creation, and configuration. Additionally, it entails an exploration of the history of Linux, its benefits, and the various Linux distributions, with a specific emphasis on Ubuntu. The scope involves a detailed walkthrough of the installation process for Ubuntu, guiding users through startup procedures, logging in, and desktop navigation. It also encompasses an in-depth exploration of the Linux file system, covering fundamental concepts such as file management, file system structure, and various file operations. Furthermore, the scope includes practical training in using command-line text editors, file searching, comparison, and manipulation, enabling users to effectively manage files and directories within the Linux environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Introduction to Virtual Machine, Creating and Configuring Virtual Machine, Linux History

PC2. Benefits of Linux, Different Flavors of Linux, Introducing Ubuntu

PC3. Installing Ubuntu: Starting Up, Logging in, Exploring the Desktop, Ubuntu Basics

PC4. Browsing the File System, Understanding File System Concept, Managing Files: Real and Virtual Files, Mounting, File Searches, File Size, File Space Viewing, Text Files Viewing, Using a Command , Line Text Editor, Creating Files, Searching through Files, Comparing Text Files, Copying, Moving, Managing Files

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Virtual Machines: Knowledge of virtualization technology and software tools for creating and configuring virtual machines. Linux History and Benefits: Understanding the origins, development, and advantages of the Linux operating system over other operating systems.
- KU2.** Linux Distributions: Awareness of various Linux distributions, their features, and use cases, with a specific focus on Ubuntu.

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- KU3.** Ubuntu Installation and Basics: Familiarity with the installation process and basic operations within the Ubuntu Linux environment, including desktop navigation and file system management.
- KU4.** File System Operations: Understanding the structure and management of file systems in Linux, including file browsing, searching, size estimation, and text file manipulation using command-line text editors. File Management: Knowledge of file management operations such as copying, moving, and comparing files, as well as handling real and virtual files and mounting external file systems.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
- The proficient execution of the outlined tasks necessitates adept organizational skills, including effective time management for efficient handling of tasks such as virtual machine configuration, Linux installation, and file management within the Ubuntu environment. Furthermore, meticulous attention to detail is crucial during file system navigation, file searches, and text file manipulation, ensuring accurate and precise data handling. Additionally, proficient task prioritization and problem-solving abilities are essential for addressing any challenges that may arise during the installation process and for maintaining an organized approach to data and file management operations.
- GS2.** • Professional Skills:
- Proficiency in the aforementioned tasks requires a combination of professional skills, including a strong grasp of virtualization technologies and Linux operating systems, effective problem-solving abilities to address technical challenges during virtual machine configuration and Linux installation, and meticulous attention to detail for accurate file system management and file manipulation. Additionally, effective communication skills are essential for documenting procedures and troubleshooting steps, while adaptability and continuous learning ensure the ability to stay updated with the latest developments in virtualization and Linux environments.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	15	-	-
PC1. Introduction to Virtual Machine, Creating and Configuring Virtual Machine, Linux History	15	15	-	-
PC2. Benefits of Linux, Different Flavors of Linux, Introducing Ubuntu	-	-	-	-
PC3. Installing Ubuntu: Starting Up, Logging in, Exploring the Desktop, Ubuntu Basics	-	-	-	-
PC4. Browsing the File System, Understanding File System Concept, Managing Files: Real and Virtual Files, Mounting, File Searches, File Size, File Space Viewing, Text Files Viewing, Using a Command , Line Text Editor, Creating Files, Searching through Files, Comparing Text Files, Copying, Moving, Managing Files	-	-	-	-
NOS Total	15	15	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1104
NOS Name	Manage data in Open source tool.
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4.5
Credits	1
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1106: Visualize data graphically--

Description

The installation process for the Hadoop framework begins with downloading the latest version from the official Apache Hadoop website and ensuring that the system meets the necessary hardware and software requirements, including the appropriate Java Development Kit (JDK) version. Similarly, for Java programming, one must first install the latest JDK, set up the necessary environmental variables, and verify the installation by running simple Java programs through the command line. Furthermore, an integrated development environment (IDE) like Eclipse or IntelliJ IDEA can be used for more sophisticated Java programming, where the necessary libraries and plugins need to be installed for seamless development and debugging.

Scope

The scope covers the following :

- The above process description outlines the initial steps involved in the installation of the Hadoop framework and Java programming. While it provides a comprehensive understanding of the essential setup procedures, the scope extends further to encompass advanced configuration options, performance tuning, and troubleshooting in the case of Hadoop, and delves into comprehensive Java programming aspects, including object-oriented programming, data structures, algorithms, and the development of complex applications for various domains, thereby facilitating the creation of robust and scalable big data solutions and versatile software applications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Installation of Hadoop, Java concepts, OOPS concepts, Looping techniques, Methods, Method Overloading, Method Over-riding, Arrays, Java Programming
- PC2.** HDFS, Architecture of Hadoop, Configuring Hadoop Cluster & Hadoop Commands.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Installation of Hadoop, Java Concepts: Object-Oriented Programming (OOP) Concepts, Looping Techniques, Methods, Method Overloading, and Method Overriding, Arrays in Java
- KU2.** HDFS (Hadoop Distributed File System): HDFS is the primary storage system used by Hadoop applications. It is designed to store very large files across multiple machines in a reliable and efficient manner, providing high-throughput access to application data. It achieves reliability by replicating the data across multiple hosts, and it is highly fault-tolerant. Architecture of Hadoop and Configuring Hadoop Cluster

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** General skills include proficiency in system setup and configuration, understanding of environmental variables, and experience in working with command-line interfaces. Professional skills involve advanced knowledge of distributed systems, troubleshooting abilities, and familiarity with big data processing tools. General skills comprise a solid grasp of basic programming principles, data structures, and algorithms. Professional skills encompass the ability to design and develop complex application
- GS2.** General skills involve a strong understanding of object-oriented principles, encapsulation, and inheritance. Professional skills include proficiency in designing modular and scalable software architectures, implementing design patterns, and ensuring code reusability and maintainability. General skills entail the ability to define and call methods in Java. Professional skills include expertise in designing robust and reusable code structures, understanding the nuances of method overloading.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	32	14	-	-
PC1. Installation of Hadoop, Java concepts, OOPS concepts, Looping techniques, Methods, Method Overloading, Method Over-riding, Arrays, Java Programming	-	-	-	-
PC2. HDFS, Architecture of Hadoop, Configuring Hadoop Cluster & Hadoop Commands.	-	-	-	-
NOS Total	32	14	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1106
NOS Name	Visualize data graphically--
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1106: Visualize data graphically.

Description

Introduction to Tableau involves understanding its data visualization capabilities and functionalities. Connecting Tableau to various data sources such as Excel, CSV, and databases allows for seamless data integration. Analyzing data using Tableau's robust features, including formatting options, sorting capabilities, and diverse charting tools, enables users to gain valuable insights and present data in a visually appealing and comprehensive manner.

Scope

The scope covers the following :

- The scope encompasses a comprehensive understanding of Tableau, focusing on its data visualization capabilities and the process of connecting to diverse data sources such as Excel, CSV, and databases. It includes extensive training in data analysis through Tableau's advanced features, encompassing formatting techniques, sorting functionalities, and various charting tools, enabling users to effectively transform complex datasets into insightful visual representations for informed decision-making and data presentation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Familiarization with Tableau's interface and its capabilities for data visualization and analysis. Learning the process of linking Tableau to Excel and CSV files for seamless data integration and analysis
- PC2.** Understanding how to establish connections between Tableau and various databases to access and analyze data directly from these sources, Exploring Tableau's data analysis tools and functionalities for dissecting and understanding complex datasets
- PC3.** Mastering formatting techniques within Tableau to enhance the visual appeal of data visualizations and dashboards. Understanding the sorting capabilities in Tableau to arrange data in a meaningful and logical order for effective analysis. Utilizing Tableau's diverse charting tools to create various types of visual representations, including bar charts, line graphs, and pie charts, for comprehensive data presentation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Introduction to Tableau and Connecting to Excel, CSV Text Files
- KU2.** Connecting to Databases and its analysis
- KU3.** Formatting, Sorting and Charts

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

GS1. • Organizational Skills:

- The effective management of the tasks outlined requires a set of crucial organizational skills, including proficient time management for efficient utilization of time during the learning process and the establishment of connections to various data sources. Moreover, meticulous task prioritization is essential for understanding the importance of each step, from data connection to analysis and visualization, while maintaining a systematic approach to data organization within the Tableau platform. Additionally, attention to detail plays a key role in ensuring the accurate representation of data through appropriate formatting, sorting, and chart selection, fostering a coherent and insightful data visualization process

GS2. • Professional Skills:

- Proficiency in the tasks outlined demands a combination of professional skills, including a comprehensive understanding of data visualization principles and the ability to effectively communicate complex data insights. Proficient technical expertise in Tableau and data connection management is essential for streamlined data integration and analysis. Critical thinking skills are crucial for in-depth data analysis and the selection of appropriate data visualization tools, while attention to detail ensures accurate and insightful data presentation. Effective problem-solving abilities play a vital role in overcoming any technical challenges, fostering a seamless and efficient data analysis and visualization process.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	40	-	-
PC1. Familiarization with Tableau's interface and its capabilities for data visualization and analysis. Learning the process of linking Tableau to Excel and CSV files for seamless data integration and analysis	20	40	-	-
PC2. Understanding how to establish connections between Tableau and various databases to access and analyze data directly from these sources, Exploring Tableau's data analysis tools and functionalities for dissecting and understanding complex datasets	-	-	-	-
PC3. Mastering formatting techniques within Tableau to enhance the visual appeal of data visualizations and dashboards. Understanding the sorting capabilities in Tableau to arrange data in a meaningful and logical order for effective analysis. Utilizing Tableau's diverse charting tools to create various types of visual representations, including bar charts, line graphs, and pie charts, for comprehensive data presentation.	-	-	-	-
NOS Total	20	40	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1106
NOS Name	Visualize data graphically.
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4.5
Credits	2
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1117: Installation of Hadoop Framework and Java Programming--

Description

Big Data concepts revolve around the management and analysis of large, complex datasets, encompassing structured, semi-structured, and unstructured data. Analyzing Big Data is crucial as it enables organizations to extract valuable insights, patterns, and trends, facilitating data-driven decision-making and enhancing operational efficiency. Within the realm of Business Intelligence, Big Data plays a pivotal role by providing comprehensive and real-time analytics, enabling businesses to gain a competitive edge through informed strategic planning, market analysis, and customer insights. MapReduce and Hive, as integral components of the Hadoop ecosystem, facilitate the processing and analysis of vast datasets.

Scope

The scope covers the following :

- The importance Big Data analysis and the roles it plays in Business Intelligence and decision-making encompasses a comprehensive understanding of the challenges and opportunities associated with managing large and diverse datasets, along with the utilization of advanced analytics tools like MapReduce and Hive for effective data processing, querying, and extraction of actionable insights, thus facilitating informed business strategies, optimized operational processes, and improved customer experiences through data-driven decision-making and predictive modeling.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Performance criteria for Big Data concepts involve the efficient handling of large and diverse datasets with a focus on scalability, data integrity, and real-time processing capabilities, ensuring high throughput, low latency, and fault tolerance. The analysis of Big Data is imperative as it aids in identifying market trends, customer preferences, and operational inefficiencies, facilitating improved decision-making, strategic planning, and targeted marketing campaigns
- PC2.** In the context of Business Intelligence, Big Data plays a crucial role by enabling organizations to gain a comprehensive understanding of market dynamics, customer behavior, and competitive landscapes, thereby fostering data-driven insights for effective resource allocation and risk management. Analysis of data using tools like MapReduce and Hive provides the means to process extensive datasets, perform complex data transformations, and generate valuable business insights.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding Big Data Concepts involves comprehending the management and analysis of vast and complex data sets, comprising structured, semi-structured, and unstructured data, with a focus on data volume, variety, velocity, and veracity. Analyzing Big Data is essential for extracting meaningful insights, identifying patterns, and making data-driven decisions that contribute to operational efficiency, innovation, and competitive advantage.

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- KU2.** Within Business Intelligence, Big Data plays a critical role in providing a holistic view of business operations, market trends, and customer behavior, thereby facilitating informed strategic planning, risk management, and targeted marketing initiatives. Analysis of data using MapReduce and Hive, fundamental components of the Hadoop ecosystem, enables the parallel processing and querying of large data sets, allowing for efficient data transformation and extraction of actionable business insights

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** The understanding of Big Data Concepts requires a strong grasp of data management principles, data analysis techniques, and familiarity with tools and technologies for handling large datasets. Professional skills in this domain include proficiency in data mining and analytics, experience with Big Data frameworks and technologies such as Hadoop, Spark, and NoSQL databases, as well as expertise in data visualization and interpretation to derive actionable insights from complex data sets.
- GS2.** Understanding of statistical methods, machine learning algorithms, and the ability to communicate complex technical concepts to nontechnical stakeholders are essential for leveraging Big Data effectively in business intelligence and decision-making processes. Proficiency in programming languages commonly used in data analysis, such as Python, R, and SQL, is also highly valuable in this field.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	37	17	-	-
PC1. Performance criteria for Big Data concepts involve the efficient handling of large and diverse datasets with a focus on scalability, data integrity, and real-time processing capabilities, ensuring high throughput, low latency, and fault tolerance. The analysis of Big Data is imperative as it aids in identifying market trends, customer preferences, and operational inefficiencies, facilitating improved decision-making, strategic planning, and targeted marketing campaigns	-	-	-	-
PC2. In the context of Business Intelligence, Big Data plays a crucial role by enabling organizations to gain a comprehensive understanding of market dynamics, customer behavior, and competitive landscapes, thereby fostering data-driven insights for effective resource allocation and risk management. Analysis of data using tools like MapReduce and Hive provides the means to process extensive datasets, perform complex data transformations, and generate valuable business insights.	-	-	-	-
NOS Total	37	17	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1117
NOS Name	Installation of Hadoop Framework and Java Programming--
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1118: Manage big data using hadoop--

Description

Advanced Java refers to the use of advanced features and functionalities in the Java programming language, including but not limited to Applets and Swings, which are used to create interactive graphical user interfaces (GUIs) for web applications and desktop software. Front-end Java Database Connectivity (JDBC) allows Java applications to interact with various databases, with the JDBC-ODBC Bridge facilitating communication between Java and ODBC databases. Creating JDBC drivers and utilizing the DSN Driver Manager enable the establishment of connections to databases.

Scope

The scope covers the following :

- The scope of the aforementioned topics involves an in-depth understanding and practical application of Advanced Java concepts, including the development of interactive and visually appealing user interfaces using Applets and Swings. It also encompasses comprehensive knowledge of front-end Java Database Connectivity (JDBC) techniques, such as the utilization of the JDBC-ODBC Bridge, creating JDBC drivers, and working with the DSN Driver Manager for seamless integration with ODBC databases.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** The performance criteria for the topics of Advanced Java, Applets, Swings, and Front-end Java Database Connectivity (JDBC) involve several key aspects. First, a solid understanding of Advanced Java is crucial for developing complex, efficient, and maintainable applications, with the ability to implement various advanced features effectively. Proficiency in developing Applets and Swings requires creating visually appealing and user-friendly graphical user interfaces with optimal performance and r
- PC2.** Expertise in utilizing JDBC-ODBC Bridge and DSN Driver Manager should result in seamless and stable communication between Java applications and ODBC databases, with a focus on data integrity and security. Moreover, proficient use of Connection, Statement, and Result Set interfaces should ensure fast and accurate data retrieval and manipulation, with careful consideration given to resource management and optimization.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Graphical user interfaces (GUIs) using Applets and Swings, along with a deep understanding of their underlying functionalities and event handling mechanisms. Additionally, a strong comprehension of front-end Java Database Connectivity (JDBC) is essential, encompassing skills in establishing connections to various databases, executing SQL queries, and managing data using JDBC drivers, Connection, Statement, and Result Set interfaces. Familiarity with the JDBC-ODBC Bridge, DSN Driver Manager.

Qualification Pack

KU2. Integration of the URL statement with Applets is essential for developing dynamic and database-driven Java applications with efficient data retrieval and manipulation capabilities. A comprehensive understanding of these topics is vital for building robust, efficient, and user-friendly Java applications with seamless database integration and interactive front-end functionalities.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** General skills include a solid foundation in programming concepts, an understanding of object-oriented programming (OOP) principles, proficiency in Java programming, and familiarity with graphical user interface (GUI) development. Moreover, a strong grasp of database fundamentals, SQL, and data management principles is crucial for effective database connectivity.
- GS2.** Professional skills encompass expertise in Advanced Java development, including the ability to design and implement complex applications, proficiency in GUI development using Applets and Swings, and in-depth knowledge of JDBC for seamless database connectivity. Additionally, strong problem-solving abilities, attention to detail, and effective communication skills are essential for collaborating with teams, understanding client requirements, and delivering efficient and reliable solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	7	-	-
PC1. The performance criteria for the topics of Advanced Java, Applets, Swings, and Front-end Java Database Connectivity (JDBC) involve several key aspects. First, a solid understanding of Advanced Java is crucial for developing complex, efficient, and maintainable applications, with the ability to implement various advanced features effectively. Proficiency in developing Applets and Swings requires creating visually appealing and user-friendly graphical user interfaces with optimal performance and r	-	-	-	-
PC2. Expertise in utilizing JDBC-ODBC Bridge and DSN Driver Manager should result in seamless and stable communication between Java applications and ODBC databases, with a focus on data integrity and security. Moreover, proficient use of Connection, Statement, and Result Set interfaces should ensure fast and accurate data retrieval and manipulation, with careful consideration given to resource management and optimization.	-	-	-	-
NOS Total	15	7	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1118
NOS Name	Manage big data using hadoop--
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1119: Implementation of use cases of Data Analytics--

Description

Introduction to database , advantages of DBMS, Concept of keys: candidate key, primary key, alternate key, Foreign key, Fundamental integrity rules: entity integrity, referential integrity. Entity-relationship model .ER-diagram, SQL, Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations. Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations.

Scope

The scope covers the following :

- It encompasses the foundational understanding of databases, the benefits of Database Management Systems (DBMS), and an in-depth exploration of key concepts including candidate keys, primary keys, alternate keys, foreign keys, entity integrity, and referential integrity. Additionally, it involves the application of the Entity-Relationship model for database design, the mastery of SQL for efficient data management, and a holistic view of Knowledge Discovery in Databases (KDD), data mining techniques, and the architecture of data warehouses. The scope further extends to encompass the methodologies of migrating data to data warehouses, the pivotal process of data cleaning, and the application of aggregation operations to ensure data accuracy and integrity for effective decision-making and analysis.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding the fundamentals of databases, their types, and their importance in information management systems.
- PC2.** Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.
- PC3.** Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.
- PC4.**
 - Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets. Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis.
 - Migrating Data to Data

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Individual learn about Concepts of Database and Its Advantages
- KU2.** Concept of Keys, Fundamental Integrity Rules, Entity-Relationship Model (ER Diagram):
- KU3.** Mastering the Structured Query Language (SQL), Knowledge Discovery in Databases (KDD),
- KU4.** Learning concepts of Data Mining and Data Warehouse

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
 - Time Management: Efficiently managing tasks and adhering to project timelines, especially during data migration and cleaning operations. Resource Management: Effectively utilizing available tools and resources for database management and data mining processes. Data Organization: Structuring data in a systematic manner to ensure efficient access and retrieval during analysis and reporting.
 - Project Planning: Strategically planning and implementing data warehouse initiatives, taking into account data migration, cleaning, and aggregation operations. Detail Orientation: Paying close attention to data details to maintain accuracy and integrity throughout the data management and analysis processes.
- GS2.** • Professional Skills:
 - Technical Proficiency: Demonstrating an advanced understanding of database management systems, SQL querying, and data mining techniques for proficient handling and analysis of large datasets.
 - Problem-Solving: Resolving complex database-related issues and challenges during data migration, cleaning, and aggregation, ensuring optimal data quality and integrity.
 - Communication Skills: Articulating complex technical concepts and insights to both technical and non-technical stakeholders in a clear and concise manner. Collaboration: Working effectively with multidisciplinary teams, including data analysts, IT professionals, and stakeholders, to ensure seamless data management and analysis processes. Critical Thinking: Evaluating data trends and patterns critically to make informed decisions during data mining and knowledge discovery processes.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	16	7	-	-
PC1. Understanding the fundamentals of databases, their types, and their importance in information management systems.	-	-	-	-
PC2. Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.	-	-	-	-
PC3. Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.	-	-	-	-
PC4. - Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets. - Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis. - Migrating Data to Data	-	-	-	-
NOS Total	16	7	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1119
NOS Name	Implementation of use cases of Data Analytics--
Sector	Information Technology Sector
Sub-Sector	Data Science & Engineering
Occupation	Data Analytics
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria.

The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, Internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NSQC Approved || National Institute of Electronics and Information Technology (NIELIT)

Qualification Pack

NIELIT HQS.

- The assessor assigns roll number.
- The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
- An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
- Pass percentage would be 50% marks in each component.
- Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
- For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
- There would be no exemption for any paper/module for candidates having similar qualifications or skills.
- The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	9
NIE/SSC/N1114.Implementation of Data Analysis Project--	-	-	-	30	30	8

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N1101.Perform basic calculation using spreadsheet--	23	10	-	-	33	9
NIE/ITS/N1102.Manage structured data--	24	10	-	-	34	10
NIE/ITS/N1103.Analyse data using spreadsheet tool--	35	10	-	-	45	13
NIE/ITS/N1104.Manage data in Open source tool--	18	15	-	-	33	10
NIE/ITS/N1106.Visualize data graphically--	32	14	-	-	46	13
NIE/ITS/N1117.Installation of Hadoop Framework and Java Programming--	37	17	-	-	54	16
NIE/ITS/N1118.Manage big data using hadoop--	15	7	-	-	22	6
NIE/ITS/N1119.Implementation of use cases of Data Analytics--	16	7	-	-	23	6
Total	220	120	-	30	370	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.